KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4301 | 4 Credits: 0 | Program Core | Source-grounded

Web Scripting Languages

- To understand the principles of creating an effective web page.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4301 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4301.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	4301
Course title	Web Scripting Languages
Semester	4 Credits: 0
Course category	Program Core
Periods per week	3 (L: 3 T: 0 P: 0) Periods per semester: 45
Periods per semester	45
Credits	0
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

28 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To understand the principles of creating an effective web page.
- ● To understand the concepts of JavaScript.
- ● To implement client side interface through DOM, JQuery & AJAX.

Course outcomes

CO	Official outcome	Study level
CO1	web, client server architecture and formulation of web 10 Understand pages using HTML. Illustrate the concepts of JavaScript and validate web	See official syllabus
CO2	11 Understand pages using JavaScript.	See official syllabus
CO3	Apply DOM to add functionality to web pages. 10 Applying. Use Ajax, JQuery & AngularJS to enhance the	See official syllabus
CO4	12 Applying. functionality of web pages..	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3 3
CO2	CO2 3 3 3
CO3	CO3 3 3 3
CO4	CO4 3 3 3 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	architecture and formulation of web pages using HTML.	8	As specified
Module 2	: Illustrate the concepts of JavaScript and validate web pages using JavaScript.	7	As specified
Module 3	: Apply DOM to add functionality to web pages.	5	As specified
Module 4	: Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.	8	As specified

MODULE 1

architecture and formulation of web pages using HTML.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: architecture and formulation of web pages using HTML.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and

1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding world web addresses (ip, domain, url), summarize client server architecture, web browser and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 1 Understanding web server communication.

1 Understanding web server communication. is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding web server communication. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding web server communication. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding web server communication. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 1 Understanding web server communication. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▾ Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML,

Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define xml, summarize xml with example. 1 understanding define html,show basic structure of html, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML,

Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize head, title and metadata in html 1 understanding summarize heading tags in html. summarize paragraph and line break tags in html, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-□□ Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ **Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML,**

Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize lists in html, summarize tables in 1 understanding html summarize image tag and its attributes in html, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes,

Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize hyperlinks in html, summarize form tag 2 understanding html. summarize form elements in html (text boxes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-□□ Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ 1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned

1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding radio buttons, check boxes, text area and buttons.) develop a web page using all the above-mentioned should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ 2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks –Lists – Tables-Image Tag and Attributes– Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons.

2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks –Lists – Tables-Image Tag and Attributes– Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks –Lists – Tables-Image Tag and Attributes– Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying html elements. addresses (ip, domain, url) - client server architecture-web browser and web sever communication, xml. describe html, basic structure of html page, basic mark-up: -headings - paragraphs - line breaks –lists – tables-image tag and attributes– hyperlinks- form, form elements in html: - text boxes, radio buttons, check boxes, text area and buttons. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks –Lists – Tables-Image Tag and Attributes– Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks –Lists – Tables-Image Tag and Attributes– Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

: Illustrate the concepts of JavaScript and validate web pages using JavaScript.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Illustrate the concepts of JavaScript and validate web pages using JavaScript.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ **webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments,**

webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, webpages more interactive, examples of javascript in 1 understanding browser. basic javascript instructions: statements, comments, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 1 Understanding variable, data types.

1 Understanding variable, data types. is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding variable, data types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding variable, data types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding variable, data types. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 1 Understanding variable, data types. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function,

JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function, is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, javascript arrays, expressions, operators. 1 understanding functions methods: function, anonymous function, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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▼ Global objects: string, number, math, date. 1 Understanding Decision making and Loops: decision making: if statement, if…else statement, switch statement, loops:

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▼ 3 Applying key loop concepts, for loops, while loops, do while loops. Define JavaScript – How JavaScript makes the webpages more interactive - examples of JavaScript in browser, Basic JavaScript instructions: statements, comments, variable, data types, arrays, expressions, operators. Functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, Global objects: string, number, math, date. Decision making and Loops: decision making: if statement, if…else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops.

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▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

: Apply DOM to add functionality to web pages.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Apply DOM to add functionality to web pages.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing

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Concept and explanation

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Study points

- Define or identify 2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, 2 understanding (dom), dom tree as a model of a web page. dom tree - working with dom tree accessing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Topic study guide

Aspect	What to prepare
Definition/identity	What listeners, the event object, event delegation, user 2 Applying. interface events, event bubbling. a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ listeners, the event object, event delegation, user 2 Applying. interface events, event bubbling. a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

: Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding works? Handling Ajax request and response, data format:

1 Understanding works? Handling Ajax request and response, data format: is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding works? Handling Ajax request and response, data format: using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding works? handling ajax request and response, data format: should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 1 Understanding works? Handling Ajax request and response, data format: means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 1 Understanding works? Handling Ajax request and response, data format: [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 1 Applying. XML. JSON – Introduction to JSON, Working with JSON

1 Applying. XML. JSON – Introduction to JSON, Working with JSON is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying. XML. JSON – Introduction to JSON, Working with JSON using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying. xml. json – introduction to json, working with json should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying. XML. JSON – Introduction to JSON, Working with JSON means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-□□ 1 Applying. XML. JSON – Introduction to JSON, Working with JSON □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery

2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying data, loading html & json with ajax. jquery : what is jquery ?, a basic jquery should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 4-□□ 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements,

2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding example, why use jquery ? jquery operations - finding elements, jquery selection, getting element content, updating elements, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 4-□□ 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ 2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular

2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying changing content, inserting elements, adding new content, getting and setting attributes. angular js: what is angular js?, why use angular should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 4-□□ 2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 1 Understanding JS?, How Angular JS works?.

1 Understanding JS?, How Angular JS works?. is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding JS?, How Angular JS works?. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding js?, how angular js works?. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding JS?, How Angular JS works?. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-□□ 1 Understanding JS?, How Angular JS works?. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▾ Angular Expression, Filters, Directives,Controllers.

Angular Expression, Filters, Directives,Controllers. is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Angular Expression, Filters, Directives,Controllers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, angular expression, filters, directives,controllers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Angular Expression, Filters, Directives,Controllers. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Angular Expression, Filters, Directives,Controllers. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ **Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes.Introduction to Angular JS, Angular Expression, Filters, Directives,Controllers. Angular JS modules & Forms.(no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, “Internet and World Wide Web How To Program”, 5/E, Pearson Education, 2012. R1 Jon Duckett , “HTML and CSS: Design and Build Websites”, Wiley R2 Jon Duckett , “JavaScript and JQuery : Interactive Front–End Web Development”, Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda”ng – book: The Complete Guide to Angular Paperback - 6”, February 2018. Online Resources Sl.No Website Link 1 https://www.w3schools.com/html/default.asp 2 https://www.tutorialspoint.com/html5/index.htm 3 https://www.w3schools.com/js/default.asp 4 https://www.w3schools.com/angular**

Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes.Introduction to Angular JS, Angular Expression, Filters, Directives,Controllers. Angular JS modules & Forms.(no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, “Internet and World Wide Web How To Program”, 5/E, Pearson Education, 2012. R1 Jon Duckett , “HTML and CSS: Design and Build Websites”, Wiley R2 Jon Duckett , “JavaScript and JQuery : Interactive Front–End Web Development”, Wiley. R3 Felipe Coury, Ari Lerner, Carlos

Taborda”ng – book: The Complete Guide to Angular Paperback - 6”, February 2018. Online Resources Sl.No Website Link 1 https://www.w3schools.com/html/default.asp 2 https://www.tutorialspoint.com/html5/index.htm 3 https://www.w3schools.com/js/default.asp 4 https://www.w3schools.com/angular is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes.Introduction to Angular JS, Angular Expression, Filters, Directives,Controllers. Angular JS modules & Forms.(no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, “Internet and World Wide Web How To Program”, 5/E, Pearson Education, 2012. R1 Jon Duckett , “HTML and CSS: Design and Build Websites”, Wiley R2 Jon Duckett , “JavaScript and JQuery : Interactive Front–End Web Development”, Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda”ng – book: The Complete Guide to Angular Paperback - 6”, February 2018. Online Resources Sl.No Website Link 1 https://www.w3schools.com/html/default.asp 2 https://www.tutorialspoint.com/html5/index.htm 3

https://www.w3schools.com/js/default.asp 4 https://www.w3schools.com/angular using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarization of angular js modules & forms. 2 applying introduction to ajax, handling ajax request and response, data format: xml, introduction to json, working with json data, loading html & json with ajax, introduction to jquery, jquery operations - finding elements, jquery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes.introduction to angular js, angular expression, filters, directives,controllers. angular js modules & forms.(no detailed study required.) text / reference t/r book title/author t1 paul j. deitel, harvey m. deitel, abbey deitel, “internet and world wide web how to program”, 5/e, pearson education, 2012. r1 jon duckett , “html and css: design and build websites”, wiley r2 jon duckett , “javascript and jquery : interactive front–end web development”, wiley. r3 felipe coury, ari lerner, carlos taborda”ng – book: the complete guide to angular paperback - 6”, february 2018. online resources sl.no website link 1 https://www.w3schools.com/html/default.asp 2 https://www.tutorialspoint.com/html5/index.htm 3 https://www.w3schools.com/js/default.asp 4 https://www.w3schools.com/angular should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content,

Aspect	What to prepare
	getting and setting attributes.Introduction to Angular JS, Angular Expression, Filters, Directives,Controllers. Angular JS modules & Forms.(no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, “Internet and World Wide Web How To Program”, 5/E, Pearson Education, 2012. R1 Jon Duckett , “HTML and CSS: Design and Build Websites”, Wiley R2 Jon Duckett , “JavaScript and JQuery : Interactive Front–End Web Development”, Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda”ng – book: The Complete Guide to Angular Paperback - 6”, February 2018. Online Resources Sl.No Website Link 1 https://www.w3schools.com/html/default.asp 2 https://www.tutorialspoint.com/html5/index.htm 3 https://www.w3schools.com/js/default.asp 4 https://www.w3schools.com/angular means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes.Introduction to Angular JS, Angular Expression, Filters, Directives,Controllers. Angular JS modules & Forms.(no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M.

Deitel, Abbey Deitel, “Internet and World Wide Web How To Program”, 5/E, Pearson Education, 2012. R1 Jon Duckett , “HTML and CSS: Design and Build Websites”, Wiley R2 Jon Duckett , “JavaScript and JQuery : Interactive Front–End Web Development”, Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda”ng – book: The Complete Guide to Angular Paperback - 6”, February 2018. Online Resources Sl.No Website Link 1 https://www.w3schools.com/html/default.asp 2 https://www.tutorialspoint.com/html5/index.htm 3 https://www.w3schools.com/js/default.asp 4 https://www.w3schools.com/angular □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps →
Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure
→ Observation → Result →
Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
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> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and. (3 marks)**

▶ **Define or explain 1 Understanding web server communication.. (3 marks)**

▶ **Define or explain Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML,. (3 marks)**

▶ **Define or explain Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML,. (3 marks)**

Module 2

▶ **Define or explain webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments,. (3 marks)**

▶ **Define or explain 1 Understanding variable, data types.. (3 marks)**

▶ **Define or explain JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function,. (3**

marks)

▶ **Define or explain 2 Understanding variable scope. Functions objects: object, this, arrays are objects,. (3 marks)**

Module 3

▶ **Define or explain 2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing. (3 marks)**

▶ **Define or explain elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing html content,. (3 marks)**

▶ **Define or explain 2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to. (3 marks)**

▶ **Define or explain bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event. (3 marks)**

Module 4

▶ **Define or explain 1 Understanding works? Handling Ajax request and response, data format:. (3 marks)**

▶ **Define or explain 1 Applying. XML. JSON – Introduction to JSON, Working with JSON. (3 marks)**

▶ **Define or explain 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery. (3 marks)**

▶ **Define or explain 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements,. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding (DOM), DOM tree as a model of a web page.**

Module 4

1-mark definition: State the meaning of **1 Understanding works? Handling Ajax request and response, data format:**.

DOM tree - working with DOM tree accessing.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- ”, Wiley
- R2 Jon Duckett , “JavaScript and JQuery : Interactive Front-End Web
- Development”, Wiley.
- R3 Felipe Coury, Ari Lerner, Carlos Taborda”ng - book: The Complete Guide to
- Angular Paperback - 6”, February 2018. Online Resources
- Sl.No Website Link https://www.w3schools.com/html/default.asp
- https://www.tutorialspoint.com/html5/index.htm https://www.w3schools.com/js/default.asp https://www.w3schools.com/angular

IN-PAGE SEARCH

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